

Decode An IR Remote For Home Automation

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This article describes how to decode a remote controller and use it for home automation, along with Arduino. An infrared (IR) remote controller used in TVs, music players and the like uses IR rays for sending information or data signals to the device. That is, the remote controller sends information in the form of IR pulses to the TV, for example, whenever a button is pressed on it. Each button is encoded with a unique code.

So, by decoding and understanding the code associated with each button, any remote controller can be used for controlling other devices or home appliances. The author's prototype is shown in Fig. 1.

Circuit and working

The project has three parts: decoding IR remote controller, programming, and making home automation circuit.

For decoding, use any TV/DVD IR remote controller.

The circuit diagram for the decoding circuit is shown in Fig. 2. It is built around Arduino Uno board (Board1) along with IR sensor TSOP 1738 wired on a bread-board. Pin out of TSOP 1738 is shown in Fig. 3.

V_{CC} , GND and OUT pins of TSOP 1738 are connected to 5V, GND and pin No. 11 of Arduino, respectively. After completing the connections, open Arduino IDE and upload the decoding code (Decode_IR.ino) to Arduino Uno board. The code consists of

Remote Keys and Codes

Remote key	Code	Purpose
VOL+	DBC8CB72	Power on home appliances
VOL-	5A33C754	Power off home appliances

a dedicated library for IR operations. Unzip IR + library + for + Arduino.zip folder and place it under Arduino/Libraries folder in Windows computer.

Point IR remote towards TSOP 1738 and press a button on the remote. Open the serial monitor to see the code of each button. The corresponding code will be displayed on the serial monitor whenever a button is pressed. Note down the code and corresponding button name for further use in home automation. Here, we used two keys for switching on and off the appliance, as shown in the table.

Programming circuit

Arduino Pro Mini board is used for home automation. Upload home_auto.ino code to Arduino Pro Mini using Arduino Uno. The programming circuit is shown in Fig. 4.

To program, remove ATmega328 microcontroller (MCU) from Arduino Uno. After setting up the connections as shown in Fig. 4, open Arduino IDE, choose Tools→Boards→Arduino Pro Mini and upload home_auto.ino code to Pro Mini board.

In home_auto.ino code, pin 13 of Arduino is declared as output. Codes DBC8CB72 and 5A33C754 are included in this source code. The two keys, Vol+ and Vol-, of the remote controller are used for turning on and turning off output pin 13 of Arduino, respec-



Fig. 1: Author's prototype

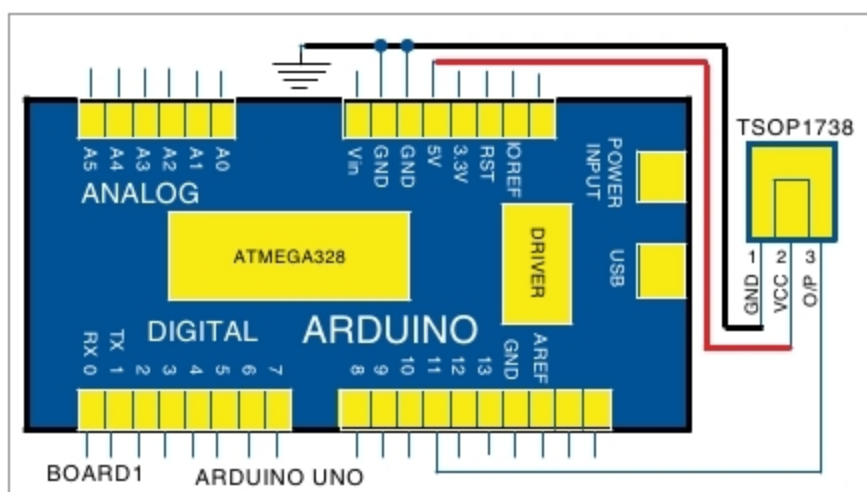


Fig. 2: Circuit diagram for IR decoding

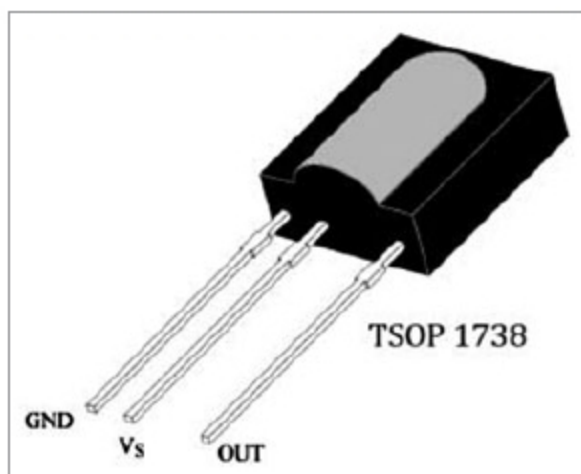


Fig. 3: TSOP 1738 pin out

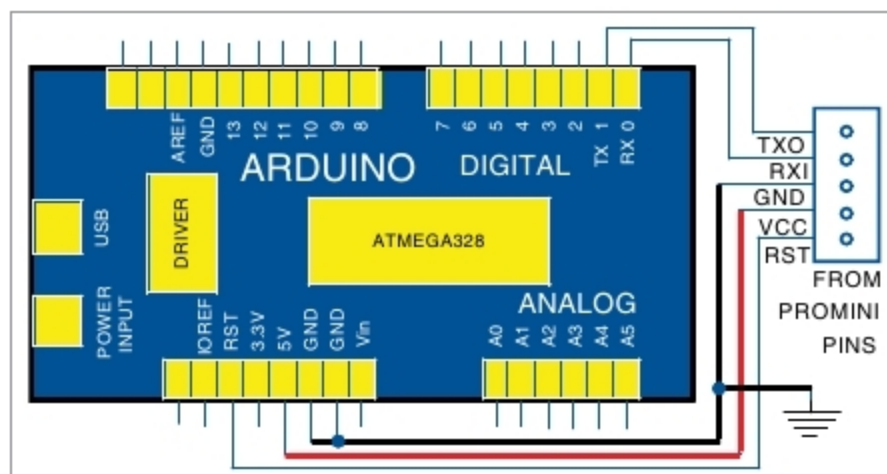


Fig. 4: Circuit diagram for programming